

Practice Geometry Test I (Version 2)

Name
Date
Course

Instructions

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show work when necessary.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Each problem is worth four points for a total of 100 points.

① If I buy new shoes, then I will go to the dance.

i) Identify the hypothesis and conclusion of the statement above.

ii) Write the inverse of the statement above.

② Find the midpoint of the segment with endpoints $A(-1, 3)$ and $B(-1, -5)$.

③ If "p" represents the hypothesis of an if-then statement and "q" represents the conclusion, write symbols for the following.

i) The original statement:

ii) The converse:

iii) The inverse:

iv) The contrapositive:

④ i) Why are if-then statements called conditional statements?

ii) Rewrite the following statement in if-then form. Write in a way that makes sense in context. It is okay to add a word if necessary.

All computers have processors.

⑤ Are the following statements biconditional? If so, rewrite in "if and only if" form.

i) If an animal is a fish, then it swims in the water.

ii) If a polygon has three sides, then it is a triangle.

⑥ Are $\overline{AB}$ and $\overline{CD}$ congruent if the coordinates of their endpoints are written below?

$A(4,0)$, $B(4,5)$, $C(1,3)$, $D(-4,3)$

⑦ If S bisects $\overline{RT}$, find x .



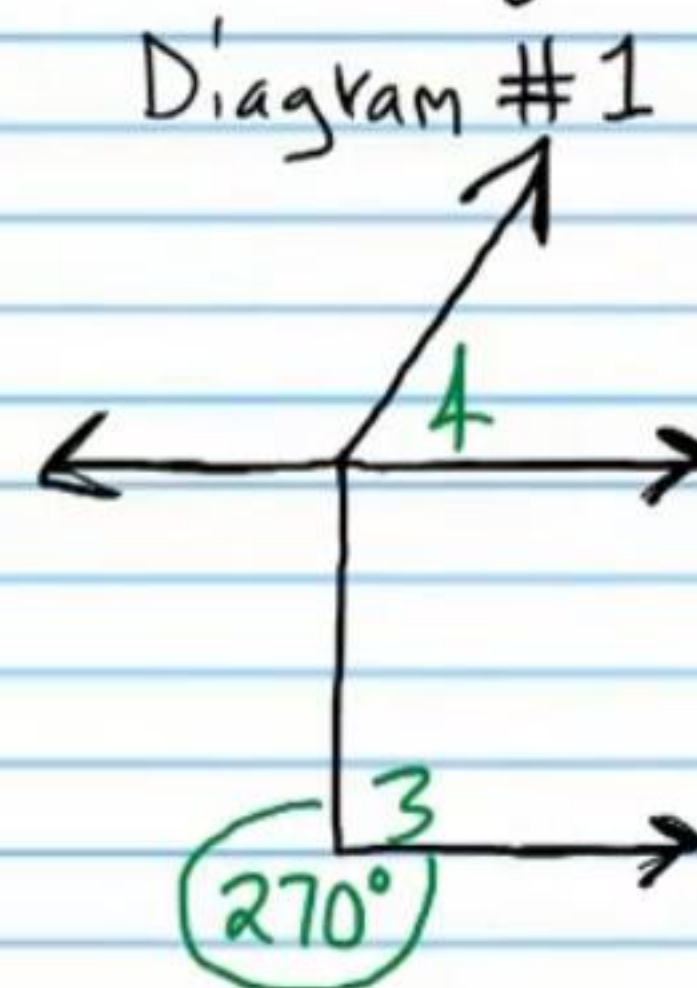
⑧ Classify the following angles (use diagram #1 for parts iii through iv).

i) 360°

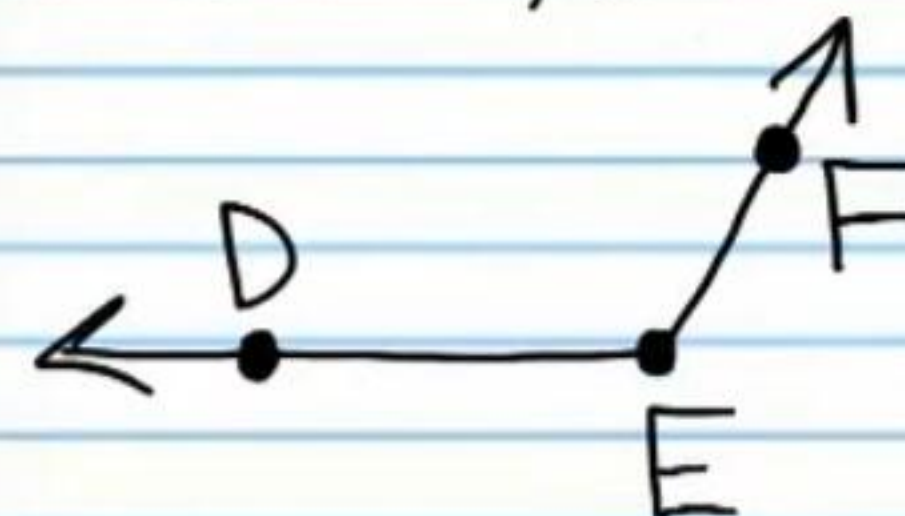
ii) 138°

iii) $\angle 4$

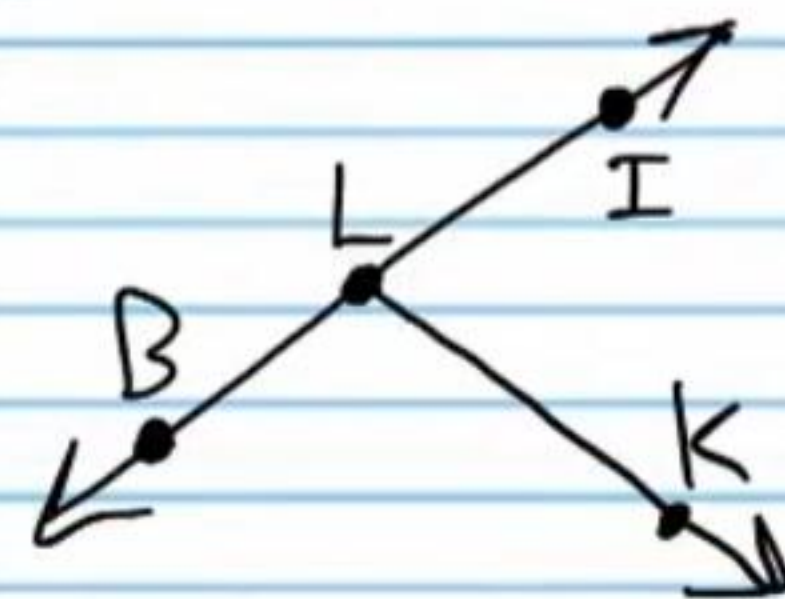
iv) $\angle 3$



⑨ Name the vertex and the sides of the angle below. Then, write three possible names for the angle.



⑩ Find $m\angle ILK$ if $m\angle BLK = 100^\circ$.



⑪ Use diagram #2 below.

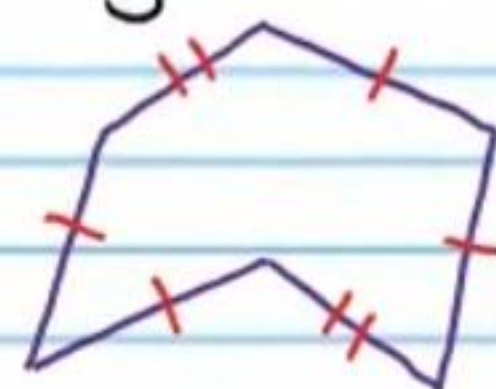
i) Name the polygon (i.e. classify it by the number of sides).

ii) Is it equilateral?

iii) Is it concave or convex?

iv) What is the name of a polygon with nine sides?

Diagram #2



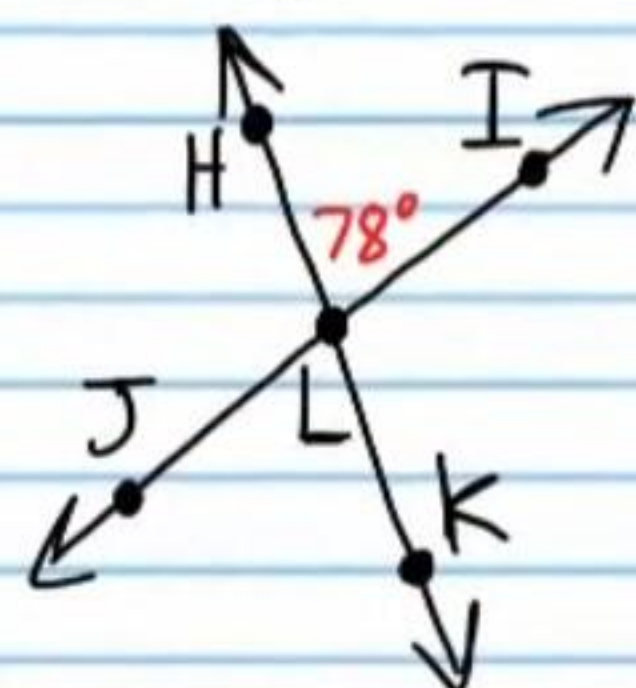
⑫ Use diagram #3 below.

i) Name two angles in the diagram that are adjacent.

ii) Name two angles that are nonadjacent.

iii) What is the measure of an angle supplementary to $\angle ILK$?

Diagram #3



iv) Name two angles that form a linear pair.

⑬ Write the steps required to construct a segment congruent to $\overline{AB}$ below.



⑭ Identify the following examples of reasoning as inductive or deductive.

i) Linsey used the Segment Addition Postulate to show that the measure of an angle is 36° .

ii) Alan used the Squeeze Theorem to prove the Fundamental Theorem of Calculus.

iii) Bill wore shorts yesterday and today.
He will wear shorts tomorrow.

iv) Lizards, dogs, and elephants have tails.
So, all animals have tails.

⑮ Make a conjecture for the next thing in the sequence.

$-36, -12, -4, \dots$

⑯ Find a counterexample for the statement below.
Every mode of transportation has four wheels.

Use diagram #4 below to answer question #'s 15-16.

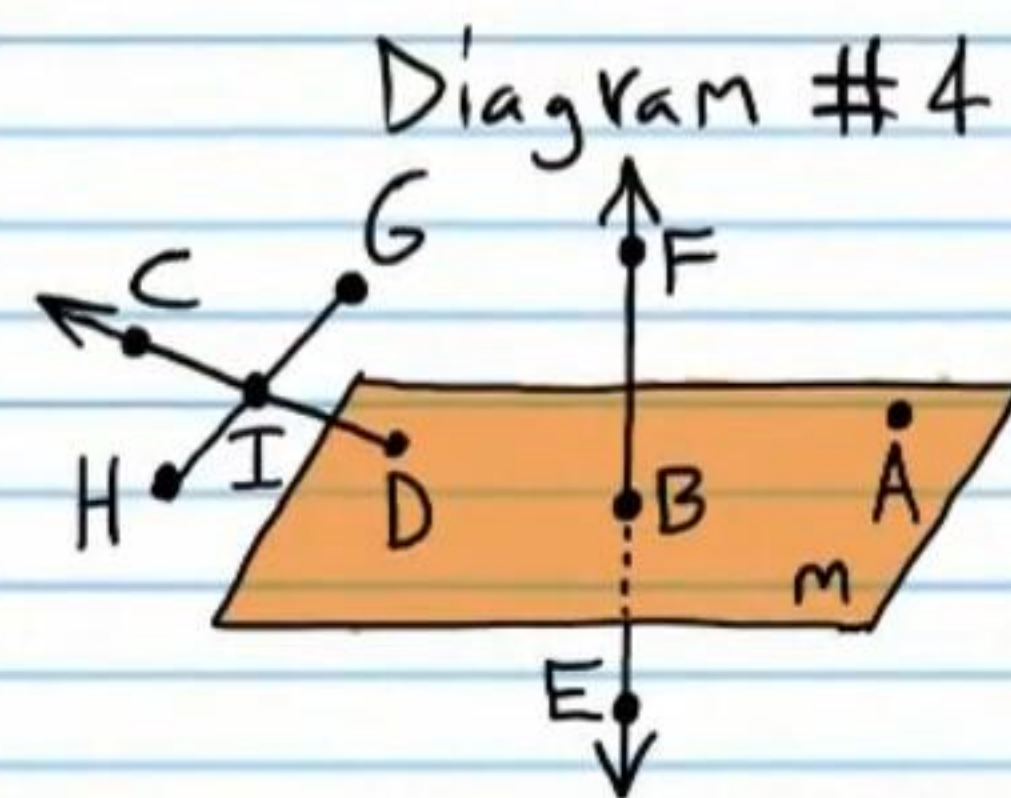
Name the type of object in the diagram that each symbol refers to (e.g. some possible answers: plane, ray).

⑰ i) E

ii) $\overleftrightarrow{FE}$

iii) DBA

iv) m



⑱ i) Name three points that are coplanar with m.

ii) What is the intersection of $\overrightarrow{DC}$ and $\overline{GH}$?

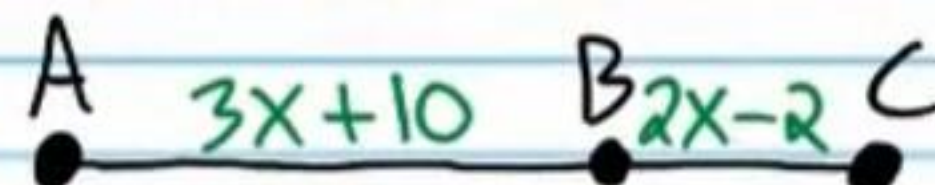
19) Determine if the Law of Detachment or the Law of Syllogism can be used for the following examples. Make a valid conclusion using one of the Laws.

i) If Amelia wins the talent show, then she will receive \$200. If she receives \$200, then she will buy a new guitar.

ii) The weather is warm today. If the weather is warm, then we can go swimming.

Write a two-column proof for each statement below.

20) If $x = 5$, then $AC = 33$.



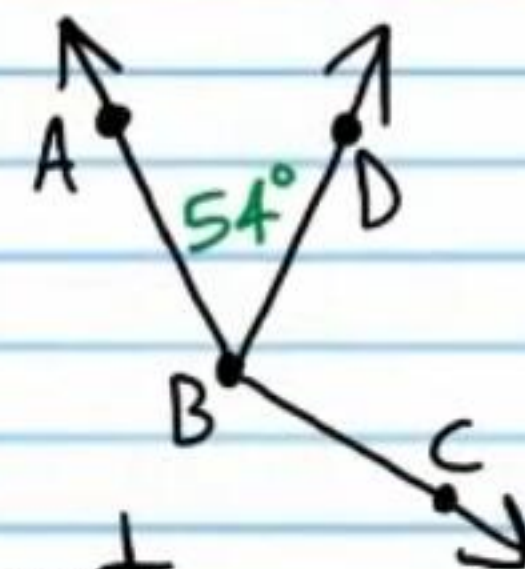
Statement	Reason
I)	
II)	
III)	
IV)	

V)

VI)

VII)

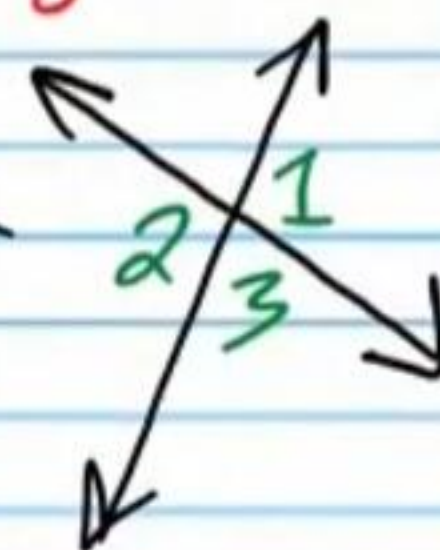
21) If $m\angle ABC = 150^\circ$ and $m\angle ABD = 54^\circ$, then $m\angle DBC = 96^\circ$.



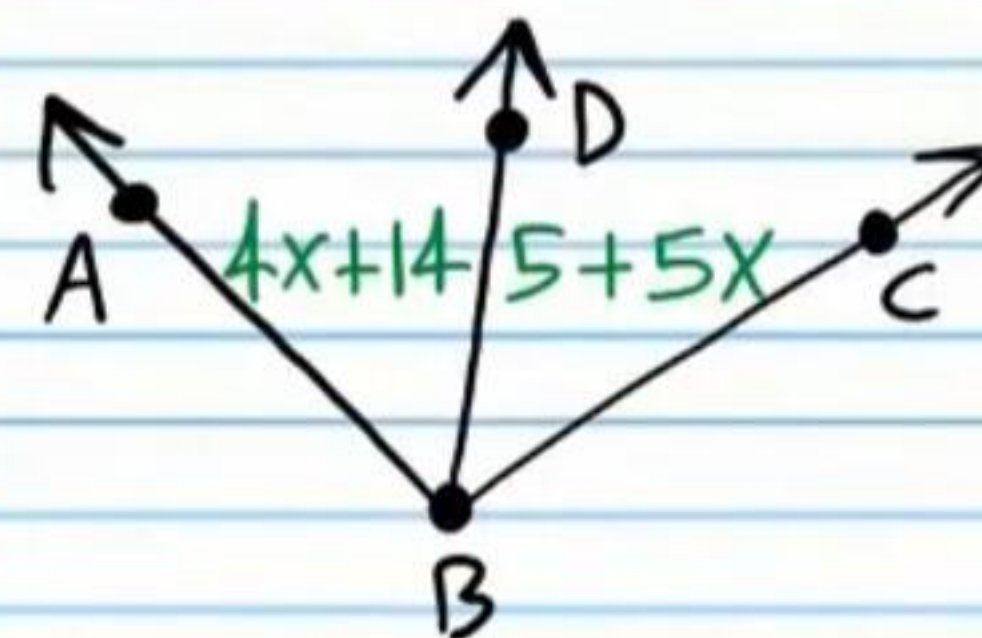
Statement	Reason
I)	
II)	
III)	
IV)	
V)	

② Use the Congruent Supplements Theorem to prove the Vertical Angles Theorem (i.e. prove that $\angle 2 \cong \angle 1$). Hint: It is given in the diagram that certain angles form linear pairs.

Statement	Reason
I)	
II)	
III)	

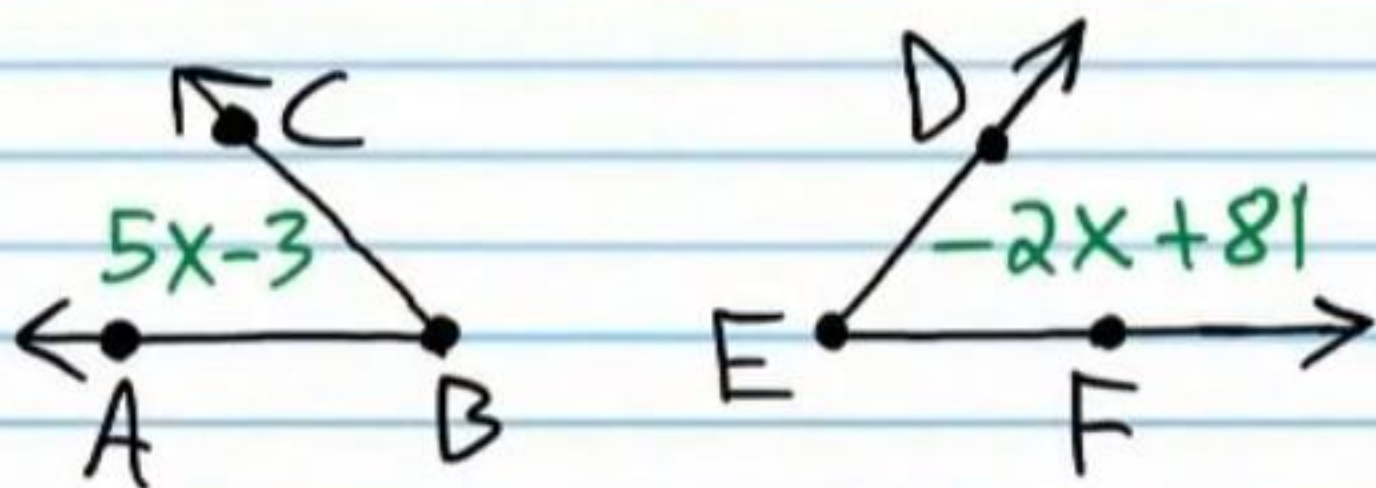


③ If $\overrightarrow{BD}$ bisects $\angle ABC$, then $x = 9$.



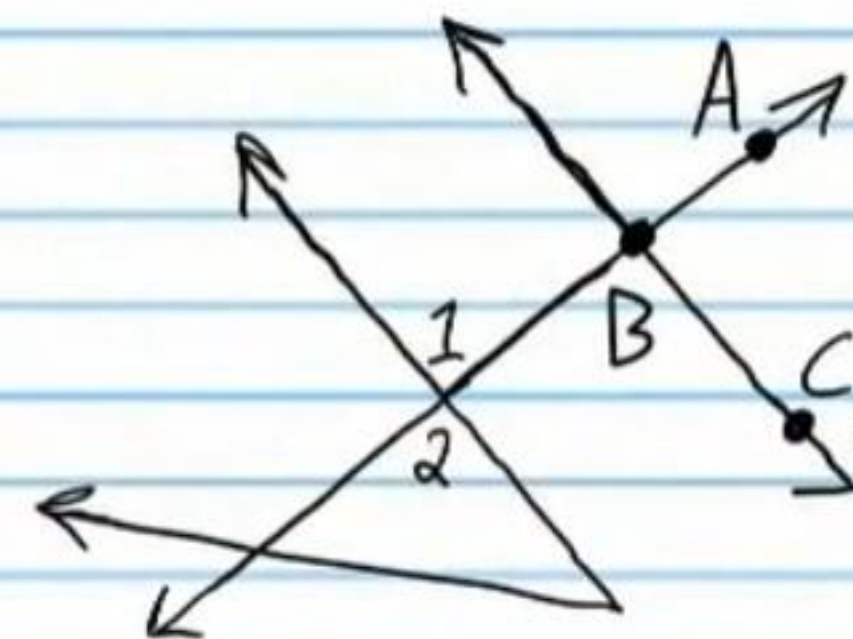
Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	

②4 If the two angles below are congruent, then $m\angle ABC = 57^\circ$



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	
VIII)	
IX)	

②5 If $m\angle ABC = m\angle 2$ and $\angle 1$ is a right angle, then $\overleftrightarrow{AB} \perp \overleftrightarrow{BC}$.



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	
VIII)	
IX)	

Practice Geometry Test I (Version 2) Answers

Grammatical variations are okay.

① i) Hypothesis: I buy new shoes.

Conclusion: I will go to the dance.

ii) If I do not buy new shoes, then I will not go to the dance.

② $(-1, -1)$

③ i) $P \rightarrow Q$

iii) $\sim P \rightarrow \sim Q$

ii) $Q \rightarrow P$

iv) $\sim Q \rightarrow \sim P$

Variations of these answers are okay.

④ i) The words that follow "if" reveal a condition required in order for the event described by the words that follow "then" to occur.

ii) If a device is a computer, then it has a processor.

⑤ i) No

ii) Yes. A polygon has three sides if and only if it is a triangle.

⑥ Yes

⑦ $x = 4$

⑧ i) Rotation angle

iii) Acute angle

ii) Obtuse angle

iv) Right angle

⑨ Vertex: E Sides: $\overrightarrow{ED}$ & $\overrightarrow{EF}$

Three ways: $\angle DEF, \angle FED, \angle E$

⑩ $m\angle ILK = 80^\circ$

⑪ i) Hexagon

ii) No

iii) Concave

iv) Nonagon

⑫ i) Possible answers: $\angle JLH$ & $\angle HLI$, $\angle HLI$ & $\angle ILK$, $\angle ILK$ & $\angle KLT$, $\angle KLT$ & $\angle JLH$.

ii) Possible answers: $\angle JLK$ & $\angle HLI$ or $\angle JLH$ & $\angle KLI$

iii) 78°

iv) Possible answers: $\angle JLH$ & $\angle HLI$, $\angle HLI$ & $\angle ILK$, $\angle ILK$ & $\angle KLT$, $\angle KLT$ & $\angle JLH$.

⑬ Draw a point and label it with a letter. Then place one side of a compass on A and extend the other side to B. With the compass remaining at this setting, place one side of the compass on the new point drawn earlier, and draw another point where the other side of the compass rests. Then connect the new points with a ruler/straightedge.

(14) i) Deductive

ii) Deductive

iii) Inductive

iv) Inductive

(15) $\boxed{-\frac{4}{3}}$

(16) Possible answers: Motorcycles, helicopters, trains, snowmobiles.

(17) i) point

ii) Line

iii) plane

iv) plane

(18) i) A, D, and B.

ii) I

(19) i) The Law of Syllogism

Conclusion: If Amelia wins the talent show, then she will buy a new guitar.

ii) The Law of Detachment

Conclusion: We can go swimming.

→ (20) There are other ways to complete this proof.

Statement	Reason
I) $AB = 3x + 10$ & $BC = 2x - 2$	Given (diagram)
II) $AC = AB + BC$	Seg. Add. Post.
III) $AC = 3x + 10 + 2x - 2$	Subs. Prop of =.
IV) $AC = 5x + 8$	Simplification.
V) $x = 5$	Given
VI) $AC = 5(5) + 8$	Subs. Prop of =.
VII) $AC = 33$	Simplification.

Statement	Reason
I) $m\angle ABC = 150^\circ$ and $m\angle ABD = 54^\circ$	Given.
II) $m\angle ABC = m\angle ABD + m\angle DBC$	Seg. Add. Post.
III) $150 = 54 + m\angle DBC$	Subs. Prop of =.
IV) $96^\circ = m\angle DBC$	Subt. Prop of =.
V) $m\angle DBC = 96^\circ$	Sym. Prop of =.

(22)

Statement	Reason
I) $\angle 2$ & $\angle 3$ form a linear pair and $\angle 3$ & $\angle 1$ form a linear pair.	Given (diagram)
II) $\angle 2$ & $\angle 3$ are supplementary angles and $\angle 3$ & $\angle 1$ are supplementary angles.	Linear Pair Post.
III) $\angle 2 \cong \angle 1$	$\cong$ Supp. Thm.

(23)

Statement	Reason
I) $\overrightarrow{BD}$ bisects $\angle ABC$.	Given.
II) $m\angle ABD = m\angle DBC$	Def. of $\angle$ bisector.
III) $m\angle ABD = 4x + 14$ and $m\angle DBC = 5 + 5x$.	Given (diagram)
IV) $4x + 14 = 5 + 5x$	Subst. Prop of $=$.
V) $4x = -9 + 5x$	Subst. Prop of $=$.
VI) $-x = -9$	Subst. Prop of $=$.
VII) $x = 9$	Div. Prop of $=$.

(24)

Statement	Reason
I) $m\angle ABC = 5x - 3$ and $m\angle DEF = -2x + 81$.	Given (diagram)
II) $\angle ABC \cong \angle DEF$	Given.
III) $m\angle ABC = m\angle DEF$	Def. of $\cong \angle$'s.
IV) $5x - 3 = -2x + 81$	Subst. Prop of $\cong$.
V) $5x = -2x + 84$	Add. Prop of $=$.
VI) $7x = 84$	Add. Prop of $=$.
VII) $x = 12$	Div. Prop of $=$.
VIII) $m\angle ABC = 5(12) - 3$	Subst. Prop of $=$.
IX) $m\angle ABC = 57^\circ$	Simplification

25 There are other variations of this proof.

Statement	Reason
I) $\angle 1$ is a right angle.	Given.
II) $m\angle 1 = 90^\circ$	Def. of a rt $\angle$.
III) $\angle 2 \cong \angle 1$	Vert. $\angle$'s Thm
IV) $m\angle 2 = m\angle 1$	Def. of $\cong \angle$'s.
V) $m\angle 2 = 90^\circ$	Trans. Prop of $=$.
VI) $m\angle ABC = m\angle 2$	Given.
VII) $m\angle ABC = 90^\circ$	Trans. Prop of $=$.
VIII) $\angle ABC$ is a right angle.	Def. of a rt. $\angle$.
IX) $\overleftrightarrow{AB} \perp \overleftrightarrow{BC}$	Def. of $\perp$.